

Introduction to Microeconomics

Lecture 13—Price Indexes

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cost-of-living index or price index Ratio of the present cost of a typical bundle of consumer goods and services compared with the cost during a base period.

ideal cost-of-living price index Cost of attaining a given level of utility at current prices relative to the cost of attaining the same utility at base-year prices.

- The ideal cost-of-living price index takes into consideration that consumers **substitute** towards relatively cheaper consumption bundles of goods and services when facing different price ratios across periods.

Laspeyres and Paasche Indexes

For a set of goods and services $\mathcal{C} = \{c_1, c_2, \dots, c_k\}$, let

- $p_{c,t}$: prevailing price of good c in period t .
- $q_{c,t}$: quantity of c consumed in period t .

Let t_n denote the current time, and t_0 denote the **base year**.

- Laspeyres Index

$$\mathcal{L}(t_0, t_n, q_{c,t_0}) = \frac{p_{c_1,t_n} q_{c_1,t_0} + p_{c_2,t_n} q_{c_2,t_0} + \dots + p_{c_k,t_n} q_{c_k,t_0}}{p_{c_1,t_0} q_{c_1,t_0} + p_{c_2,t_0} q_{c_2,t_0} + \dots + p_{c_k,t_0} q_{c_k,t_0}}$$

- Paasche Index

$$\mathcal{P}(t_0, t_n, q_{c,t_n}) = \frac{p_{c_1,t_n} q_{c_1,t_n} + p_{c_2,t_n} q_{c_2,t_n} + \dots + p_{c_k,t_n} q_{c_k,t_n}}{p_{c_1,t_0} q_{c_1,t_n} + p_{c_2,t_0} q_{c_2,t_n} + \dots + p_{c_k,t_0} q_{c_k,t_n}}$$

- In general, a price-index depends on the base year t_0 , the current year t_n , and the basket of goods q_c .

The ideal cost of living index: example

	2000	2010
Price of books	\$20.00	\$107.5
Number of books	15	
Price of food	\$2.00	\$3.00
Kg of food	100	
Expenditure	500	

Suppose the consumer has $U(F, B) = F^{0.4} B^{0.6}$

$$MRS = \frac{0.4B}{0.6F} = \frac{2}{3} \frac{B}{F}$$

In the 2000, $I_{2000} = 500$

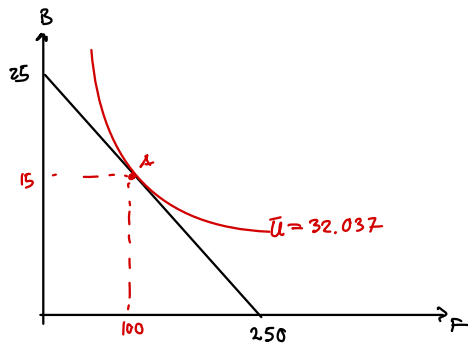
$$\begin{aligned} \max_{(F, B)} \quad & F^{0.4} B^{0.6} \\ \text{s.t.} \quad & 2F + 20B = 500 \end{aligned}$$

$$\text{Opt. Cond.} \quad \frac{2}{3} \frac{B}{F} = \frac{2}{20} \Rightarrow B = \frac{3}{20} F$$

Plug into BL: $2F + 20\left(\frac{3}{20}F\right) = 500$

$$F = 100, \quad B = \frac{3}{20} \cdot 100 = 15$$

$$A = (100, 15), \quad \bar{U} = U(A) = 100^{0.4} 15^{0.6} \approx 32.037$$



The ideal cost of living index: using the expenditure minimization problem to find the bundle in the new year

	200	2010
Price of books	\$20.00	\$107.5
Number of books	15	9
Price of good	\$2.00	\$3.00
Kg of food	100	215
Expenditure	500	1612.50

Solve the exp. min. problem

$$\min_{F, B} 3F + 107.5B$$

$$F, B \text{ subject to } F^{0.4} B^{0.6} = 32.037$$

$$\text{Opt. Condition } \frac{2}{3} \frac{B}{F} = \frac{3}{107.5} = \frac{6}{215}$$

$$B = \frac{9}{215} F$$

Plug $B = \frac{9}{215} F$ into the utility

constraint:

$$32.037 = F^{0.4} \left(\frac{9}{215} F \right)^{0.6}$$

$$= F \left(\frac{9}{215} \right)^{0.6}$$

$$F = 32.037 / \left(\frac{9}{215} \right)^{0.6} \approx 215$$

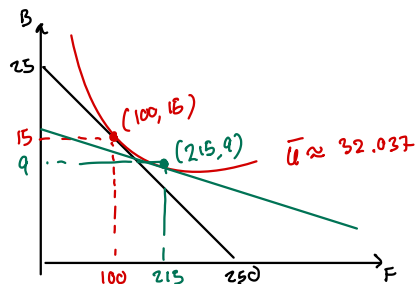
$$B = \frac{9}{215} (215) = 9$$

Bundle (215, 9)

$$\text{The cost of this bundle is: } 3(215) + 107.5(9) = 1612.50$$

The ideal cost of living index: Computing the index

	2000	2010
Price of books	\$20.00	\$107.5
Number of books	15	9
Price of good	\$2.00	\$3.00
Kg of food	100	215
Expenditure	\$500	\$1612.50



Year 2000 (100, 15)

Year 2010 (215, 9)

$$\text{Cost 2000} = 500$$

$$\text{Cost 2010} = 1612.50$$

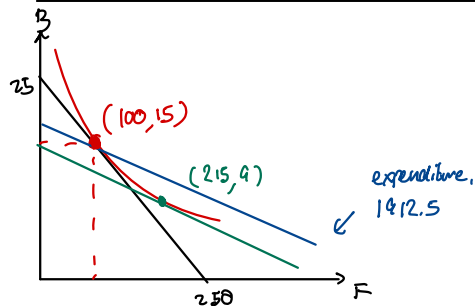
Ideal Price Index

$$I = \frac{1612.50}{500} = 3.225$$

I takes into consideration the optimal behavior of the consumer (substitute towards relatively cheaper goods).

Computing the Laspeyres Index

	2000	2010
Price of books	\$20.00	\$107.5
Number of books	15	4
Price of good	\$2.00	\$3.00
Kg of food	100	215
Expenditure	500	1412.50



Base year is 2000 (100, 15)

Cost of (100, 15) at prices from 2010:

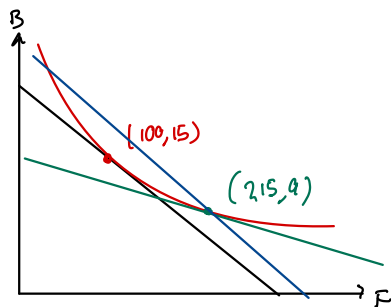
$$3(100) + (107.5)15 = 1412.5$$

$$L = \frac{\text{Cost of (100, 15) at 2010 prices}}{\text{Cost of (100, 15) at 2000 prices}} = \frac{1412.5}{500} = 3.825$$

$I \leq L \Rightarrow$ overshooting the ideal-index

Computing the Paasche Index

	200 ⁰	2010
Price of books	\$20.00	\$107.5
Number of books	15	9
Price of food	\$2.00	\$3.00
Kg of food	100	215
Expenditure	500	1612.50



Current year 2010, bundle $(215, 9)$

Compute the cost of B at current year vs base year prices (2000)

Cost $(215, 9)$ in 2010: 1612.50

Cost $(215, 9)$ in 2000: $2(215) + 20(9) = 610$

$$P = \frac{\text{Cost}(215, 9) \text{ in 2010}}{\text{Cost}(215, 9) \text{ in 2000}} = \frac{1612.50}{610} = 2.64$$

$P \leq I \Rightarrow$ undershoots the ideal index

- We have that in general:

$$\mathcal{P} \leq \mathcal{I} \leq \mathcal{L}$$

- Potential remedy is to compute **chain-weighted indexes**:
 - For the Laspeyres index:

$$\mathcal{L}(t_0, t_1, q_{c,t_0}) \times \mathcal{L}(t_1, t_2, q_{c,t_1}) \times \cdots \times \mathcal{L}(t_{n-1}, t_n, q_{c,t_{n-1}})$$

- For the Paasche index:

$$\mathcal{P}(t_0, t_1, q_{c,t_1}) \times \mathcal{P}(t_1, t_2, q_{c,t_2}) \times \cdots \times \mathcal{P}(t_{n-1}, t_n, q_{c,t_n})$$

- An example of the Laspeyres index is the CPI (Consumer Price index): Fix a base year basket and a base year and track its index across time. The CPI is used to measure inflation in all countries around the world.
- Example of a Paasche index: GDP Deflator, which is the ratio of Nominal GDP/Real GDP. As you can see this index fixes a base year and a current year basket.
 - Nominal GDP: value of goods and services at current prices.
 - Real GDP: value of goods and services at base year prices.